#_ Important [JavaScript Built-in Methods] {CheatSheet}

1. Array Methods

- forEach(): `array.forEach(element => console.log(element));`
- map(): `const squares = array.map(x => x * x);`
- filter(): `const evens = array.filter(x => x % 2 === 0);`
- reduce(): `const sum = array.reduce((acc, curr) => acc + curr, 0);`
- find(): `const found = array.find(x => x > 10);`
- indexOf(): `const index = array.indexOf(item);`
- push(): `array.push(item);`
- pop(): `const lastItem = array.pop();`
- shift(): `const firstItem = array.shift();`
- unshift(): `array.unshift(item);`
- splice(): `array.splice(startIndex, deleteCount, item);`
- slice(): `const sliced = array.slice(startIndex, endIndex);`
- concat(): `const mergedArray = array1.concat(array2);`
- join(): `const str = array.join(', ');`
- includes(): `const hasItem = array.includes(item);`
- some(): `const hasNegative = array.some(x => x < 0);`
- every(): `const allPositive = array.every(x => x > 0);`
- sort(): `array.sort((a, b) => a - b);`
- reverse(): `array.reverse();`

2. String Methods

- charAt(): `const char = str.charAt(index);`
- concat(): `const combinedStr = str1.concat(str2);`
- includes(): `const hasSubstring = str.includes(substring);`
- indexOf(): `const index = str.indexOf(substring);`
- lastIndexOf(): `const lastIndex = str.lastIndexOf(substring);`
- match(): `const matches = str.match(regex);`

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- repeat(): const repeatedStr = str.repeat(times);
- replace(): const replacedStr = str.replace(regex, newSubstr);
- search(): const index = str.search(regex);
- slice(): const slicedStr = str.slice(startIndex, endIndex);
  split(): const tokens = str.split(delimiter);
  substring(): const substring = str.substring(startIndex, endIndex);
- toLowerCase(): const lowerStr = str.toLowerCase();
- toUpperCase(): const upperStr = str.toUpperCase();
- trim(): const trimmedStr = str.trim();

3. Object Methods

- Object.keys(): const keys = Object.keys(obj);
- Object.values(): const values = Object.values(obj);
- Object.entries(): const entries = Object.entries(obj);
- Object.assign(): const newObj = Object.assign({}, obj1, obj2);
- Object.freeze(): const frozenObj = Object.freeze(obj);
- Object.isFrozen(): const isFrozen = Object.isFrozen(obj);
- Object.seal(): const sealedObj = Object.seal(obj);
- Object.isSealed(): const isSealed = Object.isSealed(obj);
- Object.create(): const newObj = Object.create(prototypeObj);
- Object.hasOwnProperty(): const hasProp = obj.hasOwnProperty(prop);

4. Number Methods

- Number.parseInt(): const intVal = Number.parseInt(string, radix);
- Number.parseFloat(): const floatVal = Number.parseFloat(string);
- Number.isNaN(): const isNaN = Number.isNaN(value);
- Number.isFinite(): const isFinite = Number.isFinite(value);
- Number.isInteger(): const isInteger = Number.isInteger(value);
- Number.toFixed(): const fixedStr = num.toFixed(digits);

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5. Math Functions

- Math.abs(): `const absolute = Math.abs(num);`
- Math.ceil(): `const roundedUp = Math.ceil(num);`
- Math.floor(): `const roundedDown = Math.floor(num);`
- Math.round(): `const rounded = Math.round(num);`
- Math.max(): `const max = Math.max(num1, num2, ...);`
- Math.min(): `const min = Math.min(num1, num2, ...);`
- Math.pow(): `const power = Math.pow(base, exponent);`

  Math.sqrt(): `const sqrt = Math.sqrt(num);`

  Math.random(): `const randomNum = Math.random();`

6. Global Functions

- parseInt(): `const intVal = parseInt(string, radix);`
- parseFloat(): `const floatVal = parseFloat(string);`
- isNaN(): `const isNaN = isNaN(value);`
- isFinite(): `const isFinite = isFinite(value);`

7. Date Methods

- Date.now(): `const now = Date.now();`
- Date.parse(): `const timestamp = Date.parse(dateString);`
- new Date(): `const date = new Date(year, month, day, ...);`

8. Regular Expression Methods

- RegExp.test(): `const matchesPattern = regex.test(string);`
- RegExp.exec(): `const match = regex.exec(string);`

9. JSON Methods

- JSON.stringify(): `const jsonString = JSON.stringify(obj);`
- JSON.parse(): `const obj = JSON.parse(jsonString);`

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10. Web APIs

- setTimeout(): `const timeoutId = setTimeout(() => { /* code */ }, delay);`
- clearTimeout(): `clearTimeout(timeoutId);`
- setInterval(): `const intervalId = setInterval(() => { /* code */ }, delay);`
- clearInterval(): `clearInterval(intervalId);`

11. Promises and Asynchronous Programming

- Promise.then(): `promise.then(value => { /* code */ });`
  Promise.catch(): `promise.catch(error => { /* code */ });`
  Promise.finally(): `promise.finally(() => { /* code */ });`
- Promise.all(): `Promise.all([promise1, promise2]).then(values => { /* code */ });`
- Promise.race(): `Promise.race([promise1, promise2]).then(value => { /* code */ });`
- async function: `async function fetchData() { const data = await fetch(url); }`

12. Console Methods

- console.log(): `console.log('message', obj);`
- console.warn(): `console.warn('warning message');` • console.error(): `console.error('error message');`
- console.info(): `console.info('info message');`
- console.clear(): `console.clear();`

13. Miscellaneous Methods

- alert(): `alert('Alert message');`
- confirm(): `const confirmed = confirm('Are you sure?');`

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- prompt(): `const userInput = prompt('Enter your name:', 'Default Name');`
- encodeURIComponent(): `const encoded = encodeURIComponent(uriComponent);`
- decodeURIComponent(): `const decoded = decodeURIComponent(encodedUriComponent);`

14. DOM Manipulation (Web-Specific)

- document.getElementById(): `const element = document.getElementById('id');`
- document.querySelector(): `const element = document.querySelector('.class');`
- document.querySelectorAll(): `const elements = document.querySelectorAll('div');`
- document.createElement(): `const div = document.createElement('div');`
  element.appendChild(): `parentElement.appendChild(childElement);`
  element.innerHTML: `element.innerHTML = '<span>New content</span>';`
- element.addEventListener(): `element.addEventListener('click', eventHandler);`
- element.style: `element.style.color = 'blue';`
- element.setAttribute(): `element.setAttribute('data-custom', 'value');`
- element.removeAttribute(): `element.removeAttribute('data-custom');`

15. Event Handling

- element.onclick: `element.onclick = function() { /* code */ };`
- element.addEventListener(): `element.addEventListener('click', eventHandler);`
- element.removeEventListener(): `element.removeEventListener('click', eventHandler);`
- event.preventDefault(): `element.addEventListener('click', event => event.preventDefault());`

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16. BOM (Browser Object Model) Methods

- window.open(): `const newWindow = window.open(url);`
- window.close(): `window.close();`
- window.moveTo(): `window.moveTo(x, y);`
- window.resizeTo(): `window.resizeTo(width, height);`

17. Web Storage API

- localStorage.setItem(): `localStorage.setItem('key', 'value');`
- localStorage.getItem(): `const value = localStorage.getItem('key');`
- sessionStorage.setItem(): `sessionStorage.setItem('key', 'value');`
- sessionStorage.getItem(): `const value = sessionStorage.getItem('key');`

18. Fetch API and XMLHttpRequest

- fetch(): `fetch(url).then(response => response.json());`
- XMLHttpRequest(): `const xhr = new XMLHttpRequest(); xhr.open('GET', url); xhr.send();`

19. History API

- history.pushState(): `history.pushState(stateObj, 'title', 'page2.html');`
- history.replaceState(): `history.replaceState(stateObj, 'title', 'page3.html');`
- history.back(): `history.back();`

20. WebSockets

- WebSocket: `const socket = new WebSocket('ws://example.com');`

21. File and FileReader API

- FileReader.readAsDataURL(): `const reader = new FileReader(); reader.readAsDataURL(file);`
- FileReader.onload: `reader.onload = function() { console.log(reader.result); };`

22. Geolocation API

- navigator.geolocation.getCurrentPosition(): `navigator.geolocation.getCurrentPosition(position => console.log(position));`
- navigator.geolocation.watchPosition(): `const watchID = navigator.geolocation.watchPosition(position => console.log(position));`

23. Performance API

- performance.now(): `const start = performance.now();`

24. Request Animation Frame

- requestAnimationFrame(): `requestAnimationFrame(timestamp => { /* animations */ });`

25. Internationalization

- Intl.DateTimeFormat(): `const formatter = new Intl.DateTimeFormat('en-US'); console.log(formatter.format(date));`
- Intl.NumberFormat(): `const numberFormatter = new Intl.NumberFormat('en-US'); console.log(numberFormatter.format(number));`

26. Canvas API

- getContext(): `const ctx = canvas.getContext('2d');`

27. Drag and Drop API

- ondragstart: `element.ondragstart = function(event) { /* code */ };`
- ondrop: `dropArea.ondrop = function(event) { /* code */ };`

28. Mutation Observer API

- MutationObserver(): `const observer = new MutationObserver(callback); observer.observe(targetNode, config);`

29. Screen API

- screen.width: `const screenWidth = screen.width;`
- screen.height: `const screenHeight = screen.height;`

30. Navigator API

- navigator.userAgent: `const userAgent = navigator.userAgent;`
- navigator.onLine: `const isOnline = navigator.onLine;`